Bio SBQ2: Answer Key

Topics 1.4 Translation · 1.5 Gene Regulation · 1.6 Cell Specialization

Read it, cover it, say it back.

Exam key words: operator · allolactose · cAMP–CAP · transcription factor · enhancer / TATA box · p53 → p21 · miRNA blocks translation · master regulator · combinational control

Data questions: always quote the numbers.

1.4 Translation

1 Amino acid sequence from 5'-GACCGC**AUG**GUGACGAAAUUUGGCCAU**UAA**-3'

Start at AUG and read in threes: AUG · GUG · ACG · AAA · UUU · GGC · CAU · **UAA (stop)**
Met – Val – Thr – Lys – Phe – Gly – His

⚠ GGC = **Gly**, not Ser. All four GG_ codons are Gly. (The chart on your practice sheet seems to misprint this box.)

2 Length of HT1 protein from 1170 bases

1170 ÷ 3 = 390 codons. Subtract 1 for the stop codon: **389 amino acids**

⚠ The stop codon codes for **no** amino acid, so always subtract 1.

3 Wild-type 796th nucleotide (mutant 795–797 = AGA, Arg; wild type = Lys)

A. Lys = AAA or AAG. The mutant codon is AGA, so only the **middle** base (796) changed, from A to G.

4 mRNA codon and amino acid for anticodon 3'-UAC-5'

The codon pairs with the anticodon (A–U, G–C): **5'-AUG-3' = Met** (the start codon)

5 Streptomycin: effect on bacteria, and why human cells are not affected

tRNA can't bind mRNA, so **translation stops** and no proteins are made. The bacteria can't grow and die.
Human ribosomes are **80S** and bacterial ones are **70S**. The structure is different, so the drug can't bind to human ribosomes.

1.5 Regulation of Gene Expression

6 Lac operon: absence vs presence of lactose

No lactose: the repressor binds the **operator** and blocks RNA polymerase, so there is **no transcription**.
Lactose present: **allolactose binds the repressor and changes its shape**. It leaves the operator, so RNA polymerase transcribes the genes.

7 Why inducible? Positive control?

Inducible: normally **OFF**, and turned **ON** by an inducer (allolactose).
Positive control: low glucose leads to high **cAMP**. cAMP binds **CAP**, CAP binds near the promoter and helps RNA polymerase bind, so there is **more** transcription.

8 **What if?** Lac operon predictions

Situation	Result	Reason
a) No repressor made	**ON**	Nothing blocks the operator, so it's on even without lactose
b) Operator mutated	**ON**	The repressor can't attach
c) Glucose high + lactose	**LOW**	The repressor is off, but low cAMP means CAP gives no boost
d) CAP can't bind cAMP	**LOW**	No positive boost to RNA polymerase

9 Phosphorylated p53, and p53 that can't bind DNA

DNA damage leads to kinases **phosphorylating p53**, which activates it. Active p53 is a **transcription factor**: it turns on **p21**, and p21 inactivates cyclin–Cdk, which **stops the cell cycle** so the DNA can be repaired.
If p53 can't bind DNA: no p21 is made, the cell keeps dividing with damaged DNA, and that can lead to **cancer**.

10 DNA methylation vs histone acetylation

	DNA methylation	Histone acetylation
Chromatin	Tight, condensed (heterochromatin)	Loose, open (euchromatin)
Genes	**OFF**	**ON**

1.6 Gene Expression & Cell Specialization

11 Why miR-P lowers enzyme but not mRNA

miR-P binds the PIG1 **mRNA after it is made** and **blocks translation**. So the mRNA stays about the same (**96**), but much less enzyme is made (**25**).

12 Normal function of the negative regulatory protein

It **reduces transcription** of PIG1. When its binding site is mutated, mRNA goes from 100 to **165** and enzyme from 100 to **158**, giving very dark purple flowers.

13 Evidence for combinational control

TF-M alone gives about **31–38**. TF-C alone gives about **8–10**. **Together they give about 88–94**, which is far more than the two added up, so they work **together**.

14 TF-M can't bind the TF-C gene

TF-C is never made, so only TF-M acts. Muscle genes stay at about **31–38 instead of about 90**, and the cells **don't fully differentiate**.

15 Same DNA, different cells: how?

Both cells have the same genes, but **different genes are switched on**. Transcription factors (TF-M and TF-C) activate muscle genes, which means different proteins, so the cells get a different shape and function.

Calc 7.8–7.9

16 $dy/dt = ky$

$y = Ce^{kt}$ (C = the starting value, y_o)

17 $dP/dt = kP(1 - P/L)$

a) Limit: **L** (the carrying capacity) b) Fastest growth at **P = L/2**